Mouhamed Salim Bousmina

Junior Software Developer | Full-Stack Development | Systems & Cybersecurity

Nabeul, Tunisia | +216 27 870 862 | bousminaselim@gmail.com | github.com/slimecoding2025 | portfolio-eight-pink-42.vercel.app

PROFESSIONAL SUMMARY

Computer science student and self-taught developer who designs, builds and documents independent software projects across web, systems and defensive security. Projects include a statically typed programming language with a compiler and bytecode VM written in Rust, a Debian-based security distribution with its own tooling, and full-stack applications in React, TypeScript, Python and Supabase. Work is tested, version-controlled and honestly scoped, with a clear separation between what is implemented and what is planned. Seeking a junior or internship role in software development, systems or application security.

EDUCATION

Licence en Technologies de l'Informatique 2026 – Present
ISET de Kélibia, Tunisia

Baccalaureate in Computer Science (Sciences Informatiques) Graduated June 2026
Lycée Rue Ali Belhouane, Nabeul, Tunisia

LANGUAGES

Arabic: Native | English: Full professional | French: Basic | German: Elementary

TECHNICAL SKILLS

Languages: Rust, TypeScript, JavaScript, Python, SQL, HTML5, CSS3, Bash

Frontend: React, Next.js, Tailwind CSS, Framer Motion, Vite, responsive design

Backend & Data: Node.js, Express, FastAPI, REST APIs, PostgreSQL, SQLite, Supabase, database design

Systems & DevOps: Linux (Debian), shell scripting, Docker, GitHub Actions CI/CD, VMware Workstation, Git, GitHub, Vercel, Render

Security: Log analysis and detection rules, MITRE ATT&CK mapping, static binary analysis, system hardening, secure input handling, digital forensics fundamentals

Engineering concepts: Compiler design, lexing and parsing, type systems, bytecode VMs, network protocol parsing, software architecture, AI API integration

Familiar / learning: PHP, MySQL, Figma, Cloudflare, distributed systems, networking fundamentals

SELECTED PROJECTS

YOMNOVA - Experimental programming language and toolchain in Rust (v0.1)
github.com/slimecoding2025/yomnova

- Designed and built an end-to-end pipeline: lexer, parser, type checker, move-based ownership checker, bytecode compiler, IR optimiser and a resource-limited virtual machine, organised as a 14-crate Rust workspace.
- Implemented structs, enums with exhaustive match, call-scoped borrowing, built-in std modules, IPv4/UDP/TCP packet parsing and an opt-in confined file API; CLI with run, check, build, exec, disasm and fmt.
- Verified by 137 Rust tests with clippy and rustfmt gates; React/Node web playground that runs the real compiler. Concurrency, TCP/UDP sockets and a distributed runtime are planned, not built.

Technologies: Rust, TypeScript, React, Node.js, Docker

AegisOS - Debian 13-based Linux distribution for defensive security and education (pre-alpha)
github.com/slimecoding2025/AegisOS

- Built the aegis CLI (Python), an apt-backed tool manager driven by a validated manifest of 138 tools, 18 categories and 12 install profiles, with health checks, reversible hardening and a local security dashboard.
- Configured a live-build UEFI ISO, built it in GitHub Actions CI (ShellCheck, pytest, .deb packaging) and booted it in VMware Workstation. Disk installation and other hypervisors are not yet verified.

Technologies: Python, Bash, Debian live-build, GitHub Actions, VMware

Traceforge - Defensive investigation workbench for Linux authentication logs
github.com/slimecoding2025/traceforge

- Parses auth.log and JSON Lines events with input limits, applies five explainable detection rules (SSH brute force, password spraying, success after failures, privileged sudo use, user probing) and maps findings to MITRE ATT&CK.
- Evidence timelines, analyst triage workflow, IP/user pseudonymisation and self-contained HTML/JSON reports; FastAPI backend with SQLite and pytest tests, React and Three.js interface. Read-only by design.

Technologies: Python, FastAPI, SQLite, React, TypeScript, Tailwind CSS

AxiomRE AI - AI-assisted reverse-engineering platform (early MVP)
github.com/slimecoding2025/axiomre

- Backend static analysis of PE/ELF binaries and source archives: hashing, format detection, strings, section entropy, imports and evidence-linked findings; AI layer rejects claims that cite unknown evidence.
- Supabase authentication, AES-256-GCM encrypted uploads, queued worker jobs and Markdown/JSON/PDF reports. AI and Supabase paths were tested against mocks only; dynamic analysis is intentionally disabled.

Technologies: Python, FastAPI, pefile, Supabase, React, TypeScript, Docker

Memora - Private notes and knowledge app (installable PWA)
github.com/slimecoding2025/Memora

- Full-text search, collections, tags, attachments, voice notes, offline capture queue and edit-conflict protection, secured with Supabase Auth, Postgres row-level security and private storage; deployed on Vercel.

Technologies: React, TypeScript, Vite, Tailwind CSS, Supabase, PostgreSQL

NEXUS ZERO - Infrastructure intelligence platform (pre-alpha foundation)
github.com/slimecoding2025/nexus-zero

- Three Rust crates: typed infrastructure graph, snapshots and diffs, and a change-impact report that lists affected resources with the evidence behind each. Includes architecture docs and a threat model; provider integrations and UI are planned.

Technologies: Rust

Other projects: VentureMind AI (Next.js 14, TypeScript, OpenRouter with JSON repair and validation); Aurazur real-estate platform (React, TypeScript, Supabase, Vercel); Bac Fel Jib AI study assistant. All on GitHub.